

Title	Authors	Year	Dataset	Application Area	Method/Model:	Metrics	Strengths:	Limitations	Research Gap	Relation to our work (Track 1 adds GNN type)
Enhanced Graph Attention Network by Integrating Transformer for Epileptic EEG Identification	Z. Xie, J. Lian, D. Wang	2025	CHB-MIT (Epilepsy) and Temple University Hospital (TUH) dataset	EEG / Epileptic Seizure Detection	GATT (Hybrid Graph Attention Network v2 + Transformer Encoder). Graph edges were defined by Pearson correlation.	Sensitivity, Specificity, Accuracy (99.40%), F1-Score, ANOVA (F-value/p-value).	Successfully captures dynamic spatial dependencies between EEG electrodes using GATv2 and temporal sequences using Transformers. Avoids data leakage by using Leave-One-Patient-Out Cross-Validation.	1. Lacks interpretability; the hybrid architecture acts as a "black box," making clinical trust difficult. 2. Highly computationally expensive, requiring massive GPU resources unsuitable for standard clinical hardware.	There is a critical need for lightweight, highly interpretable graph models in EEG analysis that provide high accuracy without requiring massive computational resources or sacrificing clinical transparency.	This paper validates the use of Pearson correlation to construct an EEG graph and strict patient-wise splitting. Our work will build upon this by using a simpler, more computationally efficient [Insert your GNN type: e.g., GraphSAGE / GCN], and we directly address their interpretability gap by applying SHAP and LIME to explain the model's predictions.
Advancement in Graph Neural Networks for EEG Signal Analysis and Application: A Review	S. M. Atoar Rahman, M. I. Khalil, H. Zhou, Y. Guo, Z. Ding, X. Gao, D. Zhang	2025	Survey of 79 EEG studies using datasets such as DEAP, SEED, CHB-MIT, PhysioNet, Bonn, TUH EEG, etc.	EEG analysis, emotion recognition, epilepsy detection, Alzheimer's disease diagnosis, motor imagery, fatigue detection, neurological disorder diagnosis	Reviews GNN, GCN, GAT, RGNN, ST-GNN, GSNN, GCNN and their variants	Summarizes Accuracy, Precision, Recall, F1-score, AUC and related metrics reported in reviewed studies.	Rigorous SLR methodology (PRISMA-style pipeline, inclusion/exclusion criteria); very current (through Dec 2024); organizes the field by both GNN variant and clinical application, useful as a map of the space	The authors themselves note the field is still largely confined to offline experiments, with little consideration given to GNN applications in real-time EEG analysis, and that current GNN architectures typically rely on static graph structures, which are not well-suited to the time-varying nature of complex EEG data, with additional noise/artifact robustness and computational cost issues	Poor generalizability across datasets, lack of interpretability ("black box" models), scalability limits on large EEG datasets, weak temporal-dynamics modeling, and near-absence of real-time/low-latency GNN pipelines	Supports using GNNs for EEG classification and motivates evaluating a GNN model on the Kaggle EEG dataset while addressing existing limitations.
Classification of epileptic seizures in EEG data based on iterative gated graph convolution network	Y. Hu, J. Liu, R. Sun, Y. Yu, Y. Sui	2024	TUSZ (Temple University Hospital EEG Seizure Corpus)	EEG / Epileptic Seizure Classification	IGGCN (Iterative Gated Graph Convolutional Network) combining Iterative Deep Graph Learning (IDGL) and Gated Graph Neural Networks (GGNN) with Focal Loss.	Accuracy, Precision, Recall (91.8%), Weighted F1-score (91.5%).	Demonstrates that dynamically optimizing the graph structure during training captures complex brain region correlations better than predefined static graphs. Effectively captures long-term temporal dependencies in EEG series.	1. Still severely affected by class imbalance; the model over-predicts the majority seizure class (CFSZ) despite using Focal Loss. 2. High computational complexity and long training times (17 hours) due to iterative graph updating, limiting real-world clinical practicality.	A model that takes 17 hours to train for a single 12-second EEG slice is not practical for real-time hospital monitoring. Your project can address this by using a static but well-justified prior graph (e.g., using Pearson correlation or Phase Locking Value) paired with a lightweight GNN (like GraphSAGE), proving that you can get comparable results with a fraction of the computational cost.	This paper justifies the superiority of GNNs over traditional CNN/LSTMs for EEG data. Our work addresses their limitations by deploying a more lightweight, static-graph [Insert your GNN type here, e.g., GraphSAGE] architecture. Furthermore, we will prioritize Macro-F1 and stratified 5-fold cross-validation to explicitly prevent the minority-class failure seen in their results.
EEG-Based Graph Neural Network Classification of Alzheimer's Disease: An Empirical Evaluation of Functional Connectivity Methods	David Klepl, Fei He, Ming Wu, Matteo De Marco, David J. Blackburn, Panagiotis G. Sarrianiannis	2022	Resting-state EEG data from Alzheimer's Disease (AD) patients and Healthy Controls (HC), represented as brain graphs using 23 EEG channels.	Alzheimer's Disease diagnosis using EEG signal classification.	Constructed EEG brain graphs using 8 Functional Connectivity (FC) measures and trained GNN models on graph structures and node features (PSD). Compared against CNN, MLP, SVM-AM, and SVM-NS baselines.	AUC, Accuracy, Sensitivity, Specificity, Precision, Recall, F1-score.	Systematically compared multiple FC graph construction methods; demonstrated that GNNs outperform CNN, SVM, and MLP baselines for EEG brain-graph classification.	Used a relatively simple GNN architecture; model interpretability remained limited ("black-box"); no single FC measure consistently outperformed the others.	Need for interpretable GNN architectures and adaptive/learnable graph construction methods for EEG classification, since the optimal FC measure remains unclear.	Demonstrates the effectiveness of GNNs for EEG classification and highlights the importance of graph construction, which is directly relevant to building a GNN model on our EEG dataset.
STGATE: Spatial-temporal graph attention network with a transformer encoder for EEG-based emotion recognition	J. Li, W. Pan, H. Huang, J. Pan, F. Wang	2023	SEED, SEED-IV, DREAMER (Emotion Recognition)	EEG / Emotion Recognition (Affective Computing)	STGATE (Spatial-Temporal Graph Attention Network + Transformer Encoder). Uses a dynamically learned adjacency matrix.	Accuracy (90.37% on SEED, 76.43% on SEED-IV, 76.35% on DREAMER), Standard Deviation.	Successfully captures both spatial brain region connections and time-frequency features. Uses strict leave-one-subject-out cross-validation to prove cross-subject generalization without data leakage.	1. Acknowledges that the model is constrained by small sample sizes, meaning the heavy Transformer architecture risks overfitting. 2. Model interpretability is limited to globally averaged topographical maps, lacking instance-level explanations for specific emotional predictions.	This paper establishes the benchmark for emotion recognition on the SEED dataset using dynamic graphs. Our work will address their limitations by deploying a more lightweight [Insert your GNN type here, e.g., GraphSAGE / GCN] to prevent overfitting on small samples. Additionally, we will introduce SHAP and LIME to provide instance-level Explainable AI (XAI), answering exactly why individual samples are classified correctly or incorrectly.	Averaging weights across the whole dataset doesn't tell us how the model makes individual decisions. It doesn't explain why Subject A was classified as "Happy" and Subject B was "Sad." Your syllabus requires you to use SHAP and LIME to explain "one correct and one wrong sample." By applying SHAP/LIME, your project will provide the instance-level transparency that this paper lacks.